Magnetic fields

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Interpret magnetic field patterns and compass direction.

SUCCESS CRITERIA - I can...

1. Label north and south poles.
2. Trace external field direction from north toward south.
3. Explain why field-line drawings are models.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “magnetic field” and “pole” to explain your model or result. Sentence starters are optional.

magnetic field: a field that influences magnets and moving charges

pole: a north or south region of a magnet

superposition: adding displacements or fields at the same place and time

01 Bring back an earlier idea

A 5 Ω resistor is in series with two parallel 6 Ω resistors across 16 V.
Find parallel equivalent, total resistance and supply current.

Word help: series = connected along one unbranched current path

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $B_{net,\,x} = B_{1,\,x} + B_{2,\,x}$; $B_{net,\,y} = B_{1,\,y} + B_{2,\,y}$.
Add field components at the same point. A compass follows the net field. Outside a bar magnet, arrows run north to south.

Worked reasoning: Outside a bar magnet, field arrows run from the north pole toward the south pole. A compass's north-seeking end follows the local field direction.

Your turn: Draw curved external field lines and two small compass arrows at different points.

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

03 Represent and interpret

Label N and S on the bar magnet.
Sketch external field arrows and the direction a nearby compass would point.

Sentence starter: The ... in my model represents ... . Its label shows ... .

N	S

Marzano check - circle: ① ② ③

04 Solve and justify

A compass between two identical magnets is observed to have no stable preferred direction at one location.
Explain what field superposition could mean there; name a limitation of that observation.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Analyze the reasoning

A student draws field lines that cross at one point. **Explain** why this is inconsistent with a single net field direction at that point.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: 1 2 3

06 Choose a method

In separate cases, an ideal blackbody temperature becomes 4 times as large and a sound frequency becomes 6 times as large at fixed sound speed. **Compare** the two wavelength changes.

Word help: wavelength = distance between corresponding points on repeated waves

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: 1 2 3

07 Exit check

If a bar magnet is cut in half, do you obtain an isolated north pole and an isolated south pole? **Explain** the usual result.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Two magnets face N toward N. **Sketch** the field qualitatively and predict the force. Then reverse one magnet and explain both changes.

Sentence starter: I predict ... . My model supports this because ... .

Marzano check - circle: 1 2 3